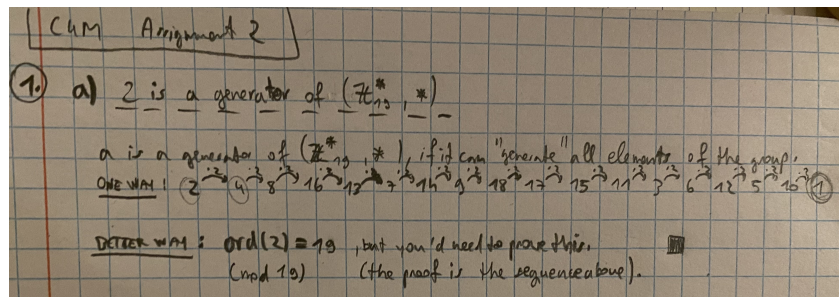


M4C 2025 - Assignment 2

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Problem 1



- (a)
- (b) Because 2 is a generator of the group $\mathbb{Z}_{19}$, every element is a power of 2 (mod 19). But if we're looking for an element which has order 18, we wouldn't find any, since the order of each element in our group has to be a divisor of 19. Since 18 and 19 are coprime, none of the elements can have the order 18.
- (c) If a group G has a prime order p (p number of elements), then the order of every element in the group has to be either 1 or the prime number p . This is because of the corollary after the Lagrange's theorem states, that the order of any element in the group has to divide the number of elements in that group. Since p is prime (it's only divisible by 1 and p), all of the orders are either 1 or p . Since 2 is the lowest prime number, we have 2 or more elements. If all the elements were of order 1, they would all be equivalent to the identity element, which would mean that we only had 1 element (contradictory with the fact that we have 2 or more elements). Therefore at least one element has order p , which means that it generates the whole group, which means the group is cyclic.

Problem 2

C4M Assignment 2

② a) 557 is prime ($\sqrt{557} \approx 23,60$)

We can see that $\frac{557}{2}, \frac{557}{3}, \frac{557}{5} \notin \mathbb{Z}$ from strategies learnt in middle school.

$557 : 7 \approx 79,57 \notin \mathbb{Z}$	Since $23 > \sqrt{557}$ and 557 not divisible by any lower prime, 557 is prime
$557 : 11 \approx 50,64 \notin \mathbb{Z}$	
$557 : 13 \approx 42,85 \notin \mathbb{Z}$	
$557 : 17 \approx 32,76 \notin \mathbb{Z}$	
$557 : 19 \approx 29,32 \notin \mathbb{Z}$	
$557 : 23 \approx 24,22 \notin \mathbb{Z}$	

(a)

Optional: If we want to check that p is prime, we can only check the first few prime numbers that are smaller (or equal) than $\sqrt{p}$. The reason for this is that if p is not prime (and $\sqrt{p} \notin \mathbb{Z}$) then p has at least 2 divisors $a, b \in \{2, 3, \dots, p-1\}$ so that $p = ab$. Both a and b cannot be larger than $\sqrt{p}$, since they would not multiply to p .

- (b) CHECK AGAIN Little Fermat's theorem doesn't help us in this case. It tells us that $x^{556} \equiv 1 \pmod{557}$ for every integer x that isn't divisible by 557.

Problem 3

(a) ...